Question Bank

Chapter-5 Magnetism and Matter

SECTION A

MCQ

Q1. When freely suspended, a magnet comes to rest in the direction

(a) North- South (b) East –West (c) South – East (d) None of the above

Q2. Magnetic lines of force always form

(a) Closed loops (b) open loops (c) both a and b (d) None of the above

Q3. The SI unit of magnetic field strength is

(a) Tesla (b) Weber (c) Tesla meter (d) Tesla meter /ampere

Q4. The SI unit of magnetic pole strength is

(a) ampere/metre^2 (b) ampere –meter (c) ampere-meter^2 (d) ampere^2 /meter

Q5. The SI unit of magnetic dipole moment is

(a) ampere –meter^2 (b) ampere –meter (c) ampere-meter^2 (d) ampere^2 /meter

Q6. Magnetic dipole moment is

(a) Scalar (b) vector (c) none

Q7. The torque acting on a magnet with magnetic dipole moment M at an angle θ with the magnetic field B is

(a) $\tau = MB\cos\theta$ (b) $\tau = MB \sin \theta$ (c) $\tau = MB \tan \theta$ (d) None

Q8. Magnetic lines of force

(a) Emanate from N- pole and enter into S- pole

(b) Emanate from S- pole and enter into N- pole

(c) Emanate from S pole to infinity

(d) Emanate from N pole to infinity

Q9. The meniscus of a liquid contained in one of the limbs of a narrow U-tube is placed between the pole-pieces of an electromagnet with meniscus in a line width the field. When the electromagnet is switched on, the liquid is seen to rise in the limb. This indicates that the liquids is

(a) Ferromagnetic (b) paramagnetic (c) Diamagnetic (d) non-magnetic

Q10. For a paramagnetic substance, the magnetic susceptibility is directly proportional to

(a) T (b) T^{2} (c) T^{0} (d) T^{-1}

Q11. The domain formation is a necessary feature of

(a) Diamagnetism (b) Paramagnetism (c) ferromagnetism (d) All of these

Q12.If a magnetic substance is kept in a magnetic field, then which of the following substances is thrown out

(a) Diamagnetism (b) Paramagnetism (c) ferromagnetism (d) All of these

Q13.Above Curie's temperature ferromagnetic substances behaves like (CBSE 2020)

(a) paramagnetic (b) diamagnetic (c) superconductor (d) no change

Q14.A permanent magnet attracts

(a) all substances (b) only ferromagnetic substances

(c) some substances and repels others (d) ferromagnetic substances and repels all others

Q15.Susceptibility is positive for

(a) paramagnetic substances (b) diamagnetic substances

(c) non- magnetic substances (d) all of the above

1 MARK QUESTIONS

Q16. What is the SI unit of magnetic dipole moment?

Q17. Can two magnetic field lines intersect?

Q18. What is the direction of magnetic dipole moment?

Q19. What is the torque acting on a bar magnet of magnetic moment M in a uniform magnetic field B?

Q20. What is the SI unit of magnetic flux density?

Q21. Are the magnetic moment of a bar magnet and its equivalent solenoid, having the same magnetic field equal?

Q22. Why do magnetic lines of force form continuous closed loops?

Q23. What is magnetic susceptibility? (CBSE 2018)

Q24. What is permeability of the material?

Q25(A). Which of the following substances are diamagnetic? (CBSE 2013)

Bi, Al, Na, Cu, Ca and Ni

(B) The susceptibility of a magnetic material is 1.9×10^{-5}. Name the type of magnetic materials it represents.

2 MARKS QUESTIONS

Q26. (a) Define magnetic field strength. (b) Give its SI unit.

Q27. Define magnetic dipole moment. Is it scalar or vector?

Q28. Give four properties of magnetic field lines.

Q29. Define intensity of magnetization of a magnetic material.

Q30. What are permanent magnets? Give examples.

Q31. Write three points of differences between diamagnetic paramagnetic and ferromagnetic-material. (CBSE 2019)

3 MARK QUESTIONS

Q32. (a) What is the name given to the curves, the tangent to which at any point gives the direction of magnetic field at that point?

(b)Can two such curves intersect each other? Justify your answer.

5 MARK QUESTIONS

Q33. (a) Derive an expression for magnetic field intensity due to a magnetic dipole at a point on its axial line.

(b)A magnetised needle of magnetic moment $4.8X10^{-2}JT^{-1}$ is placed at 30° with the direction of uniform magnetic field of $3 X 10^{-2}$ T. Calculate the torque acting on the needle. (CBSE2012)

Q34. Explain the following –

(a) Why are the field lines repelled when a diamagnetic material is placed in an external uniform magnetic field?

(b) Draw the magnetic field lines for a current carrying solenoid, when a rod made of – (i) Copper (ii) Aluminium (iii) Iron are inserted within the solenoid.

Assertion and Reason Type Questions

Select the correct answer to these questions from the codes (a), (b), (c) and (d) are as given below

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false and R is also false.

Q35.**Assertion:** Difference between an electric line and magnetic line of force is that electric lines of force are discontinuous and the magnetic field lines are continuous.

Reason: Electric lines of forces do not exist inside a charged conductor but magnetic lines exist inside a magnet.

Q36.**Assertion:** A current carrying solenoid behaves like a bar magnet.

Reason: The circular loop in which the direction of current is clockwise behaves like the South Pole and the one having anticlockwise current behaves like the North Pole.

Q37.**Assertion:** Permanent magnets retain their ferromagnetic property for a long period of time.

Reason: Steel is a diamagnetic material.

Q38**. Assertion:** When a bar magnet is hung freely it points toward geographical poles.

Reason: Magnetic field lines do not intersect.

Q39. **Assertion:** A diamagnetic specimen would move towards the weaker region of the field.

Reason: A diamagnetic specimen is repelled by a magnet.

Q40.**Assertion:** Motion of electron around a positively charged nucleus is different from the of a planet around the sun.

Reason: The force acting in both the cases is same in nature.

Q41.**Assertion:** Two parallel conducting wires carrying currents in same direction, come close to each other.

Reason: Parallel currents attract and anti-parallel currents repel.

Q42.**Assertion:** A galvanometer cannot as such be used as an ammeter to measure the current across a given section of the circuit.

Reason: For this it must be connected in series with the circuit.

Q43.**Assertion:** Magnetic lines of force form continuous closed loops whereas electric lines of force do not.

Reason: Magnetic poles always occur in pairs as North Pole and South Pole.

Q44. **Assertion:** An electron moving along the direction of magnetic field experiences no force.

Reason: The force on electron moving along the direction of magnetic field is $F = qVB \sin 0° = 0$

****

ANSWERS-SECTION A

MCQ

1. (a) North- South
2. (a) Closed loops
3. (a) Tesla
4. (b)Ampere –meter
5. (a) ampere-meter^2
6. (b) vector
7. (b) τ = MB sin θ
8. (a) Emanate from N- pole and enter into S- pole
9. (b) Paramagnetic
10. (d) T^{-1}
11. (c) ferromagnetism
12. (a) Diamagnetism
13. (a) paramagnetic
14. (b) only ferromagnetic substances
15. (a) paramagnetic substances

1 MARK

16. Ampere –metre^2 or JT^{-1}
17. No, magnetic field lines cannot intersect as at the point of intersection there are two directions of magnetic field which is not possible.
18. The direction of magnetic dipole moment is from South Pole to North Pole.
19. Torque τ =M X B = MBsinθ
20. Weber/meter^2, Tesla, Newton/ ampere -meter
21. Yes as the bar magnet and solenoid produces equal magnetic field
22. Magnetic lines of force form continuous closed loops because a magnet is always a dipole and as a result, the net magnetic flux of a magnet is always zero.
23. The measure of the magnetisation of the material is called magnetic susceptibility.
24. In electromagnetism, the measure of the resistance of a material against the formation of a magnetic field is called permeability.

25(A). Diamagnetic substances are Bi and Cu.

25(B). It represents Paramagnetic substance.

2 MARKS

26.(a) The magnetic field strength at any point in a magnetic field is defined as the force experienced by unit north pole placed at that point. b) SI unit is Tesla.

27. Magnetic dipole moment is the product of pole strength and magnetic length of the magnet.

28. (i) Magnetic field lines start from a North Pole and end on a South pole.

(ii) Magnetic field lines do not intersect each other.

(iii) Magnetic field lines are crowded where the field is stronger and farther apart where the field lines are weaker.

(iv) Magnetic field lines form closed loops.

29. The intensity of magnetization of a magnetic material is defined as the magnetic moment developed per unit volume of the material when it is placed in a magnetizing field.

30. The substances which at room temperature retain their ferromagnetism for a long time are called permanent magnets. Examples – Cobalt, Steel, Alnico and Ticonal.

31.

PROPERTIES	Diamagnetic material	Paramagnetic material	Ferromagnetic material
Effect of Magnet	Weakly repelled by a magnet.	Weakly attracted by a magnet.	Strongly attracted by a magnet
Effect of Temperature	No effect.	With the rise of temperature, it becomes a diamagnetic.	Above curie point, it becomes a paramagnetic
Permeability	Little less than unity	Little greater than unity	Very high

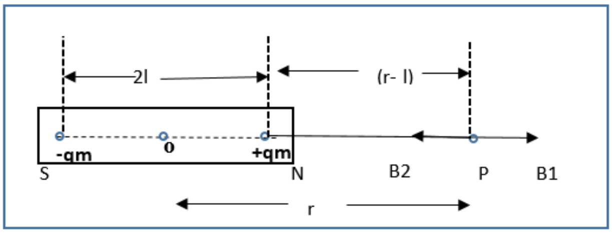
3 MARKS

32.(a) These curves are magnetic field lines. No two such lines intersect each other.

(b) Two such curves never intersect each other, if they do so, then there will be two directions at the point of intersection, which is not possible.

5 MARKS

33.(a)



Consider a magnetic dipole (or a bar magnet) SN of length 2l having South Pole at S and North Pole at N. The strength of south and north poles are – qm and + qm respectively.

Magnetic moment of magnetic dipole m = qm 2l, its direction is from S to N.

Consider a point P on the axis of magnetic dipole at a distance r from mid-point O of dipole.

The distance of point P from N-pole, r_1= (r – l).

The distance of point P from S-pole, r_2= (r + l)

Let B1 and B2 be the magnetic field intensities at point P due to north and south poles respectively. The directions of magnetic field due to North Pole is away from N-pole and due to South Pole is towards the S-pole. Therefore,

$$B_1 = \frac{\mu_0}{4\pi} \frac{q_m}{(r-l)^2}$$ from N to P and

$$B_2 = \frac{\mu_0}{4\pi} \frac{q_m}{(r+l)^2}$$ from P to S

B1>B2 and direction of resultant is from N to P and magnitude is given by

$$B = B_1 - B_2 = \frac{\mu_0}{4\pi} \frac{2(qm\ 2l)r}{(r^2-l^2)2}$$

But qm 2l =M magnetic dipole moment

$$B = \frac{\mu_0}{4\pi} \frac{2(M\ )r}{(r^2-l^2)2}$$

For a short dipole

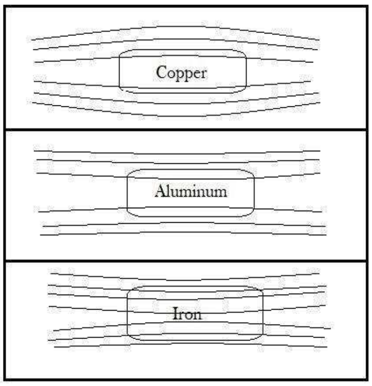
$$B = \frac{\mu_0}{4\pi} \frac{2M}{r^3}$$

(b) Torque $\tau = MB \sin\theta = 4.8\text{X}10^{-2}\text{X}3\text{X}10^{-2} \sin 30 = 7.2\text{x}10^{-4}$ Nm.

34. (a) Diamagnetic materials are the materials whose atoms do not possess any permanent magnetic dipole moment due to the presence of electrons that are paired with each other, called paired electrons. This is the reason why when a diamagnetic material is placed in an external magnetic field, the magnetic field lines are repelled.

On the application of an external magnetic field, it creates a magnetic field in the diamagnetic material opposite to the direction of the applied field which causes a repulsive force. This is the reason why when a diamagnetic material is placed in an external magnetic field, the magnetic field lines are repelled.

(b) Magnetic field lines of a current carrying solenoid: When current passed through the solenoid, it acts as bar magnet. One end of the solenoid acts as North Pole and another end acts as South Pole. The magnetic field starts at North Pole and ends at South Pole. The magnetic field lines are always parallel.



Copper:

Copper is a diamagnetic material. The magnetic field produced by the material is repulsive. Thus, the magnetic field lines diverge outward.

Aluminium:

Aluminium is a paramagnetic material. The magnetic field is strengthened around the material. Thus, the magnetic field of lines converges lightly.

Iron:

Iron is a ferromagnetic material. The magnetic field is far strong around the material. Thus, the magnetic field of lines converges heavily.

Assertion and Reason

35.(a) Since no electric lines of forces exist inside a charged body, the electric lines of force only travel from positive to negative charge and are discontinuous. Secondly, magnetic lines of force travel from north to South Pole and inside the magnet they are from South Pole to North Pole hence continuous.

36.(a) A solenoid is a type of electromagnet formed by a helical coil of wire whose length is substantially greater than its diameter, its two ends can be visualised as two coils.

37.(c) Permanent magnets retain their ferromagnetic property for a long period of time and steel is a paramagnetic material.

38.(b) It is the property of a magnet to rest in a geographical north and south pole and another property of magnetic field is that magnetic field lines do not intersect.

39.(a) In the case of diamagnetic substances, the magnetic moments of atoms and the orbital magnetic moments have been oriented in such a manner that the vector sum of an atom's magnetic moment becomes zero. An external magnetic field can repel them weakly.
40.(d) Neil Bohr proposed a model, which is familiar as a planetary model of atoms. In Bohr's model, the neutrons and protons occupy a dense central region called the nucleus, and electrons orbit the nucleus much like planets orbiting the sun. Electrons are negatively charged and the nucleus is positively charged. Force in the former case is electrostatic force but in later case it is gravitational force.
41.(a) By using the right hand thumb rule, the direction of the magnetic field can be determined then by using Fleming's right hand rule the direction of force comes towards each other.
42.(d) As the galvanometer is used to check the current flow direction and the magnitude of the direct current. That's why the resistance of the galvanometer is nearly zero. This is somewhere similar to ammeter but both are different devices. Ammeter can only show us the current magnitude not the direction. A Galvanometer's needle can fluctuate in two directions whereas an ammeter's needle can only show one side deflection.
43.(b) Electric lines of forces do not exist inside a charged body, the magnetic lines of force travel from north to south pole and inside the magnet they are from south pole to north pole hence continuous.
44.(a) If the particle is moving along the direction of magnetic field then $\theta = 0°$ hence force becomes zero.

****

SECTION B

Q1. Torque acting on a coil is maximum when the coil is placed

(a) Parallel to the magnetic field

(b) at an angle of 30° to the magnetic field

(c) at any position in a uniform magnetic field

(d) Perpendicular to the magnetic field

Q2. A long current carrying solenoid produces a magnetic field B along its axis. If the current is halved and number of turns/cm is doubled, the magnetic field becomes

(a) B/2 (b) B (c) 8B (d) 2B

Q3. The magnetic field due to a bar magnet at an equatorial point is B, its magnetic field at an axial point at the same distance is

(a) 2B (b) B/2 (c) 4B (d) B

Q 4. Current do not flow between two charged particles when connected if they have same

(a) Capacitance (b) potential (c) charge (d) none of these

Q5. Which of the following is responsible for the earth's magnetic field?

(a) Convective currents in earth's core

(b) Divertive current in earth's core.

(c) Rotational motion of earth.

(d) Translational motion of earth.

Q6. Magnetic field can be produced by

(a) a stationary charge (b) a moving charge (c) a changing electric field (d) Both b and c

Q7. Which of the following statements is true about magnetic field intensity?

(a). Magnetic field intensity is the number of lines of force crossing per unit volume.

(b). Magnetic field intensity is the number of lines of force crossing per unit area.

(c). Magnetic field intensity is the magnetic induction force acting on a unit magnetic pole.

(d). Magnetic field intensity is the magnetic moment per unit volume.

Q8. The meniscus of a liquid contained in one of the limbs of a narrow U-tube is placed between the pole-pieces of an electromagnet with meniscus in a line with the field. When the electromagnet is switched on, the liquid is seen to rise in the limb. This indicates that the liquids is

(a)Ferromagnetic (b)Paramagnetic (c) Diamagnetic (d) non-magnetic

Q9.In a permanent magnet at room temperature

(a) Magnetic moment of each molecule is zero
(b) The individual molecules have non-zero magnetic moment which are all perfectly aligned.
(c) Domains are partially aligned.
(d) Domains are all perfectly aligned

Q10.Ferromagnetic Material used in transformer must have

(a) Low Permeability & High Hysteresis Loss
(b) High Permeability & Low Hysteresis Loss
(c) High Permeability & High Hysteresis Loss
(d) Low Permeability & Low Hysteresis Loss

Q11.If the magnetising field on a ferromagnetic material is increased, if permeability is

(a) Decreased (b) Increased (c) unaffected (d)May be decreased or increased

Q12.Electromagnets are made of Soft Iron because Soft Iron has

(a) Small Susceptibility & Small Retentivity
(b) Large Susceptibility & Small Retentivity
(c) Large Permeability & Large Retentivity
(d) Small Permeability & Large Retentivity.

1 MARK QUESTIONS

Q13.How does the (i) pole strength ii) Magnetic moment of each part of a bar magnet change if it is cut into two equal pieces transverse to length?

Q14. A magnetic field interacts with a moving charge and not with a stationary charge. Why?

3.The susceptibility of a magnetic materials is -4.2×10^{-6}. Name the type of magnetic material it represents. (CBSE 2010)

Q15.The susceptibility of a magnetic material is 1.9×10^{-5}. Identify the types of magnetic material. (CBSE 2011)

Q16.The susceptibility of a magnetic material is -2.6×10^{-5}. Identify the types of magnetic material.

2 MARKS QUESTIONS

Q17. When is the torque acting on a dipole (a) Maximum (b) Minimum

Q18. Derive an expression for torque acting on a magnetic dipole placed in a uniform magnetic field.

Q19.Why is the core of an electromagnet made of ferromagnetic materials?

Q20.In what way is the behaviour of a diamagnetic material different from that of paramagnetic when kept in an external magnetic field.

Q21.Explain Curie's law for a paramagnetic substance.

3-MARKS QUESTIONS

Q22.A short bar magnet of magnetic moment $0.9JT^{-1}$ is placed with its axis at 30° to a uniform magnetic field. It experiences a torque of 0.063J. Calculate (i) the magnitude of the magnetic field (ii) in which orientation the bar magnet will be in stable equilibrium? (CBSE2012)

Q23. (a)Define the term magnetic susceptibility and write its relation in terms of relative magnetic permeability. (b)Two magnetic materials A and B have relative magnetic permeabilities of 0.96 and 500. Identify the magnetic materials A and B.

5-MARKS QUESTIONS

Q24. A short bar magnet of is placed with its axis at 30° to a uniform magnetic field of 0.16T. It experiences a torque 0.032J (a) Calculate the magnitude of the magnetic moment of the magnet. (b) In which orientation the magnet will be in stable equilibrium with the magnetic field? (c) Find the PE in stable equilibrium.

Assertion and Reason Type Questions

Assertion-Reason

Select the correct answer to these questions from the codes (a), (b), (c) and (d) are as given below

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false and R is also false.

Q25.**Assertion** The poles of magnet cannot be separated by breaking into two pieces.

Reason The magnetic dipole moment will be reduced to half when broken into two equal pieces.

Q26.**Assertion** When a bar magnet is kept in an external uniform magnetic field, it starts oscillating.

Reason A restoring torque acts on the dipole when kept in the magnetic field.

Q27.**Assertion** Two parallel wires carrying currents in the opposite direction, attract each other.

Reason Parallel currents repel and antiparallel currents attract.

Q28. **Assertion** Gauss's theorem is not applicable in magnetism.

Reason Magnetic monopoles do not exist.

Q29.**Assertion** Magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason There is a uniform magnetic field inside the solenoid.

Q30. **Assertion** If a charged particle is projected in a region, where B is perpendicular to velocity of projection, then the net force acting on the particle is independent of its mass.

Reason The particle is performing rectilinear motion.

Q31.**Assertion** When a charged particle moves in a region of magnetic field such that its velocity is at some acute angle with the direction of field, its trajectory is a helix.

Reason Perpendicular component of velocity causes a rotating centripetal force and the parallel component of velocity does not produce any force.

Q32. **Assertion** For a current carrying wire loop of N turns, placed in a region of a uniform magnetic field B, the torque acting on it is given by m x B.

Reason Whenever the magnetic moment m is perpendicular to B, then torque on the loop will be zero.

Q33.**Assertion** The current sensitivity of a galvanometer is the deflection of current per unit current passing through the coil.

Reason The galvanometer can be used as a detector to check if a current is flowing in the circuit.

Q34. **Assertion** Magnetic field lines always form closed loops.

Reason Moving charges or currents produce a magnetic field.

****

SECTION B- ANSWERS

1. (d) Perpendicular to the magnetic field
2. (b) B
3. (a) 2B
4. (b) potential
5. (a) Convective currents in earth's core
6. (d) Both b and c
7. (c). Magnetic field intensity is the magnetic induction force acting on a unit magnetic pole.
8. (b) Paramagnetic
9. (c) Domains are partially aligned.
10. (b) High Permeability & Low Hysteresis Loss
11. (a) Decreased
12. (b) Large Susceptibility & Small Retentivity.

1-MARK

13.When a bar magnet of magnetic moment (M = m2L) is cut into two equal pieces transverse to its length, its (i) the pole strength remains unchanged ii) the magnetic moment is reduced to half.

14. A moving charge experiences a force in an external magnetic field due to the interaction of two magnetic fields, one which is produced due to motion of the charge and other due to the external magnetic field, while a stationary charge does not experience any such force.

15.Susceptibility of material is negative, so given material is diamagnetic.

16.The material having positive and small susceptibility is paramagnetic material.

The material having negative and small susceptibility is diamagnetic material.

2-MARKS

17.Torque acting on a dipole is a) Maximum when θ is 90° and b) Minimum when θ is 0° or 180

18. Force on N pole = qm B along B

Force on S pole = qm B opposite to B

This give rise to a torque given by

τ = Force X perpendicular distance

= qmB X 2lsinθ= MBsinθ

19.Ferromagnetic material has a high permeability. So on passing current through windings, it gains sufficient magnetism immediately.

20.A diamagnetic specimen would move towards the weaker region of the field while a paramagnetic specimen would move towards the stronger region.

21.Curie law states that magnetization of paramagnetic substance is directly proportional to the external magnetic field applied. But as the substance is heated, it's magnetization is inversely proportional to the temperature of substance.

Thus, magnetization of paramagnetic substance M = $\frac{CB}{T}$

where C is curie constant, B is external magnetic field and T is the temperature of substance.

3 MARKS

22. (i) B= $\frac{\tau}{Msin\theta}$ = $\frac{0.063}{0.9sin30}$ =0.14T

(ii) When θ = 0° U= -MBcos 0 = -MB

P.E of the bar magnet will be minimum and it will be in stable equilibrium when it is parallel to the magnetic field.

23. (a) Magnetic Susceptibility: It is defined as the ratio of the magnetisation M to the magnetising field intensity H. It is denoted by χ_m .

$$\chi_m = \frac{M}{H}$$

Magnetic susceptibility in terms of magnetic permeability.

$$\chi_m = \mu_r - 1$$

(b) A is a diamagnetic material, B is a ferromagnetic material.

5 MARKS

24. (a) Torque) τ = MB sin θ

M = τ/B sin θ

= $\frac{0.032}{0.16}$ sin 30

= 0.4JT^{-1}

(b) For stable equilibrium θ = 0°

(c) PE is given by U = -MBcos0 = -0.4X 0.16 = -0.064J

Assertion and Reason

25.(b) The magnetic dipole moment will be reduced to half when broken into two equal pieces and every atom behaves like a dipole so the dipole of a magnet cannot be separated.

26.(a) When a bar magnet is placed in a uniform magnetic field. Then the bar magnet will experience only torque and no force, and this torque on the bar magnet will be acting on both ends, and will be equal but opposite in direction.

27.(d) Parallel currents attract and antiparallel currents repel.

28.(a) Gauss's law of magnetism is different from that for electrostatics because electric charges do not necessarily exist in pairs but magnetic monopoles do not exist.

29.(b) $B = \mu_0 nI$, from this formula we see the dependence of B in current and inside a solenoid it is uniform.

30.(c) Force =q(V x B) it is independent of mass and if v and B are perpendicular to each other, the particle describes a circle.

31.(a) A charged particle moves in a circle when its velocity is perpendicular to the magnetic field. When it forms an acute angle with the magnetic field, it can be resolved in two components, parallel and perpendicular. The perpendicular components tend to move it in a circle, the parallel components tend to move along the magnetic field to form a helical motion of uniform radius and pitch.

32.(c) Torque, $\tau = mB\sin\theta$

Here $\theta = 90°$, $\sin 90° = 1$ so torque will be maximum.

33.(b) A galvanometer is a device that is used to detect small electric current or measure its magnitude. The current sensitivity of a galvanometer is the deflection of current per unit current passing through the coil $I_S = \frac{NAB}{k}$

34.(b) Assertion is the property of a magnet while reason is one of the sources of magnetic field.

****

SECTION C

Q1. The maximum torque acting on a rectangular coil of 2cm X 5cm, having 200 turns and carrying a current of 2A is

(a) 0.4 Nm (b) 4 Nm (c) 40 Nm (d) 0.04 Nm

Q2. A toroid of n turns, mean radius R and cross-sectional radius a carries current I. It is placed on a horizontal table taken as X-Y plane. Its magnetic moment M

(a) is non-zero and points in the Z-direction by symmetry.

(b) points along the axis of the toroid.

(c) is zero, otherwise there would be a field falling as $1/r^3$ at large distances outside the toroid.

(d) is pointing radially outwards

Q3. A long magnet is cut into two parts such that the ratio of their lengths is 2:1. What is the ratio pole strength of both the section?

(a). 1:2 (b). 2:1 (c). 4:1 (d) Equal

Q4. The susceptibility of a ferromagnetic material is χ at 27^0C . At what temperature will its susceptibility be 0.5χ.

(a) 54^0C (b) 327^0 C (c) 600^0 C (d) $237\,^0$C

Q5. A paramagnetic sample shows a net magnetism of 8 Am $^{-1}$ when placed in an external magnetic field of 0.6 T at a temperature of 4K. when the same sample is placed in an external magnetic field of 0.2 T at a temperature of 16 K, the magnetization will be

(a) $\frac{32}{3}$ Am $^{-1}$ (b) $\frac{2}{3}$ Am $^{-1}$ (c) 6 Am $^{-1}$ (d) 2.4 Am $^{-1}$

Q6. The SI unit of Magnetic Permeability is

(a) $WA^{-1}m^{-1}$

(b) $NA^{-1}m^{-1}$

(c) NA^{-2}

(d) $Both\ \ WA^{-1}m^{-1}$ and NA^{-2}

Q7.Needles N1, N2 and N3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will

(a) Attract N1 strongly but repel N2 and N3 weakly.

(b) Attract all 3 of them

(c) Attract N1 and N2 strongly but repel N3

(d) Attract N1 strongly, N2 weakly and repel N3 weakly.

1MARK QUESTIONS

Q8. When a bar magnet of magnetic moment (M = m2L) is cut into two equal pieces transverse to its length, its (i) the pole strength remains unchanged (ii) the magnetic moment is halved.

Q9. The permeability of a magnetic material is 0.9983. Name the type of magnetic materials it represents. (CBSE 2011)

Q10.Susceptibility of iron is more than that of copper. What does this statement imply?

Q11. The magnetic field lines prefer to pass through iron than air. Explain why? (CBSE 2011)

Q12. Mention two characteristic properties of the material suitable for making core of a transformer. (CBSE 2012)

2 MARK QUESTIONS

Q13.A hypothetical bar magnet (AB) is cut into two equal parts. One part is now kept over the other, so that the pole C2 is above C1. If M is the magnetic moment of the original magnet, what would be the magnetic moment of the combination, so formed?

Q14. State Gauss's law for magnetism. Explain its significance. CBSE (2019)

Q15. A iron rod of length L and magnetic moment M is bent in the form of a semicircle. Find its magnetic moment.

Q16. How will you decide whether the magnetic field at a point is due to some current carrying conductor or earth?

Q16(B). A small magnetised needle P is placed at the origin of x-y plane with its magnetic moment pointing along the y-axis. Another identical magnetised needle Q is placed in two positions, one by one.

Case 1: at (a, 0) with its magnetic moment pointing along x-axis.

Case 2: at (0, a) with its magnetic moment pointing along y-axis.

(a) In which case is the potential energy of P and Q minimum?

(b) In which case is P and Q not in equilibrium? Justify your answers. (CBSE 2023)

Q17.The magnetic susceptibility of magnesium at 300K is 1.2×10^5. At what temperature will its magnetic susceptibility become 1.44×10^5? (CBSE 2019)

Q18. Show diagrammatically the behaviour of magnetic field lines in the presence of

(i) paramagnetic and

(ii) diamagnetic substances. How does one explain this distinguishing feature? (CBSE 2020)

Q19.Two substances A and B have their relative permeabilities slightly greater and slightly less than 1 respectively. What do you conclude about A and B as far as their magnetic materials are concerned?

3 MARK QUESTIONS

Q20. A rectangular current carrying loop EFGH is kept in a uniform magnetic field as shown below in the figure.

i) What is the direction of magnetic moment of current loop?

ii) When is the torque acting on the loop (a) maximum (b) minimum?

E F

H G

Q21. Write three important properties of the magnetic field lines due to a bar magnet.

Q22. A short bar magnet placed with its axis at 30° with a uniform magnetic field of 0.1 T experiences a torque of $4X10^{-2}$J. Find the magnetic moment of the magnet.

Q23. A bar magnet of magnetic moment 6 JT^{-1} is aligned at 60 ° with uniform external magnetic field of 0.44T. Calculate the work done to align its magnetic moment i) normal to the magnetic field ii) opposite to the magnetic field.

Q24. A bar magnet of magnetic moment m and moment of inertia I (about centre perpendicular to length) is cut into two equal pieces, perpendicular to its length. Let T be the period of oscillations of the original magnet about an axis through mid-point, perpendicular to the length, in a magnetic field. What would be the similar Time Period T for each piece?

Q25. On what factors does the pole strength of a magnet depends?

Q26. Show that the time period (T) of oscillations of a freely suspended magnetic dipole of magnetic moment (m) in a uniform magnetic field (B) is given by T= $2\pi\sqrt{I/MB}$, where I is a moment of inertia of the magnetic dipole. (CBSE 2019)

Q27. For a magnetising field of intensity 2 x 10^3 Am^{-1}, aluminium at 280 K acquires intensity of magnetisation of 4.8 x 10^{-2} Am^{-1},. If the temperature of the metal is raised to 320 K, what will be the intensity of magnetisation?

5 MARK QUESTIONS

Q28. (a) Two magnets of magnetic moments m and $\sqrt{3}$m are joined to form a cross (+).The combination is suspended freely in a uniform magnetic field. In equilibrium position the magnet of magnetic moment m makes an angle θ with the field. Find θ.

(b) A coil of N turns and radius R carries a current I It is unwound and rewound to make another coil of radius R/2, current I remaining the same. Calculate the ratio of magnetic moments of the new coil and the original coil. (CBSE 2012, 2015)

Q29. A small compass needle of magnetic moment 'm' is free to turn about an axis perpendicular to the direction of uniform magnetic field 'B'. The moment of inertia of the needle about the axis is 'I'. The needle is slightly disturbed from its stable position and then released. Prove that it executes simple harmonic motion. Hence deduce the expression for its time period. (CBSE 2013)

Assertion and Reason Type Questions

Select the correct answer to these questions from the codes (a), (b), (c) and (d) are as given below

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false and R is also false.

Q30. **Assertion** Steady current is the only source of magnetic field.

Reason Only moving charge can create magnetic field.

Q31. **Assertion** A magnetic field does not interact with a stationary charge.

Reason A moving charge produces a magnetic field.

Q32**.** **Assertion** When velocity of electron is perpendicular to **B** it will perform circular motion.

Reason Magnetic force is perpendicular to velocity.

Q33. **Assertion** A beam of electrons can pass undeflected through a region of E and B.

Reason Force on moving charged particles due to magnetic field may be zero in some cases.

Q34. **Assertion** If the path of a charged particle in a region of uniform electric and magnetic field is not a circle, then its kinetic energy will not remain constant.

Reason In a combined electric and magnetic field region, a moving charge experiences a net force $F = qE + q(v \times B)$, where symbols have their usual meanings.

Q35. **Assertion** If we increase the current sensitivity of a galvanometer by increasing the number of turns, its voltage sensitivity also increases.

Reason Resistance of a wire also increases with N.

Q36. **Assertion** When a magnetic dipole is placed in a non uniform magnetic field, only a torque acts on the dipole.

Reason Force would not act on dipole if magnetic field were non uniform.

Q37. **Assertion** Galvanometer can as such be used as an ammeter to measure the value of the current in given circuit

Reason It gives a full-scale deflection for a current of the order of ampere.

Q38. **Assertion** Diamagnetic materials can exhibit magnetism.

Reason Diamagnetic materials have permanent magnetic dipole moments.

Q39. **Assertion** Paramagnetic materials can exhibit magnetism.

Reason Paramagnetic materials have permanent magnetic dipole moments.

CASE STUDY BASED QUESTION

Q40. MAGNETIC BEHAVIOUR OF MATERIALS

Before nineteenth century, scientists believed that magnetic properties were confined to a few materials like iron, cobalt and nickel. But in 1846, Curie and Faraday discovered that all the materials in the universe are magnetic to some extent. These magnetic substances are categorized into two groups. Weak magnetic materials are called diamagnetic and paramagnetic materials. Strong magnetic materials are called ferromagnetic materials. According to the modern theory of magnetism, the magnetic response of any material is due to circulating electrons in the atoms. Each such electron has a magnetic moment in a direction perpendicular to the plane of circulation. In magnetic materials all these magnetic moments due to the orbital and spin motion of all the electrons in any atom, vectorially add upto a resultant magnetic moment. The magnitude and direction of these resultant magnetic moment is responsible for the behaviour of the materials. For diamagnetic materials X is small and negative and for paramagnetic materials X is small and positive. Ferromagnetic materials have large X and are characterized by nonlinear relation between B and H.

Questions: Answer any four of the following

1. The universal (or inherent) property among all substances is
   (a) Diamagnetism
   (b) Paramagnetism
   (c) Ferromagnetism
   (d) Both (a) and (b)
2. When a bar is placed near a strong magnetic field and it is repelled, then the material of bar is
   a) Diamagnetic
   b) Ferromagnetic
   c) Paramagnetic
   d) Anti-ferromagnetic
3. Magnetic susceptibility of a diamagnetic substance
   (a) Decreases with temperature
   (b) Is not affected by temperature
   (c) Increases with temperature
   (d) First increases then decreases with temperature
4. The value of the magnetic susceptibility for a super conductor is
   (a) Zero (b) Infinity (c) +1 (d) -1
5. Which of the following is weakly repelled by a magnetic field
   (a) Iron (b) Cobalt (c) Steel (d) Copper

41. Torque & Potential energy of a Dipole

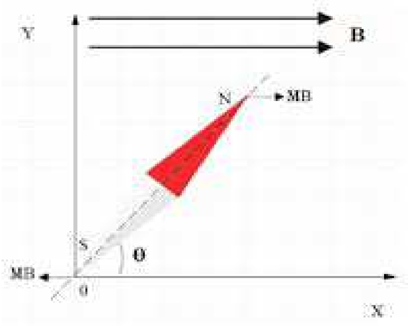
The pattern of iron fillings, i.e. the magnetic field lines give us an approximate idea of magnetic field **B**. We may at times be required to determine the magnitude of B accurately. This is done by placing a small compass needle of known magnetic moment **m** and moment of inertia **I** and allowing it to oscillate in the magnetic field. This arrangement is shown in the fig.

The torque on the needle is $\tau = m \times B$

In magnitude $\tau = mB\sin\theta$

The magnetic potential energy is given by

$U_m = -m.B$



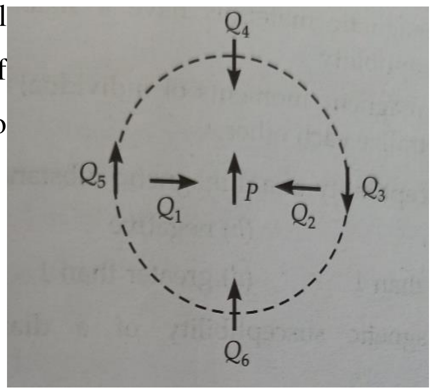
1.A bar magnet is held perpendicular to a uniform magnetic field. If the couple acting on a magnet is to be halved, by rotating it, the angle by which it is to be rotated is

(a) 30^0 (b) 60^0 (c) 45^0 (d) 90^0

2. The couple acting on a magnet of length 10 cm and pole strength 15 Am, kept in a field of V = 2×10^{-5} T at an angle 30^0 is

(a) 1.5×10^{-5} Nm (b) 1.5×10^{-3} Nm © 1.5×10^{-2} Nm (d) 1.5×10^{-6} Nm

3. The figure below shows the various positions of small magnetised needles P and Q. The arrows show the direction of their magnetic moments. Which configuration corresponds to lowest potential energy among all the configurations shown.



(a) PQ_3 (b) PQ_4 © PQ_5

(d)PQ_6

4. Two wires of the same length are shaped into a square and a circle if they carry the same current, ratio of magnetic moments is

(a) $2 : \pi$ (b) $\pi : 2$ (c) $\pi : 4$ (d) $4 : \pi$

SOURCE BASED QUESTION

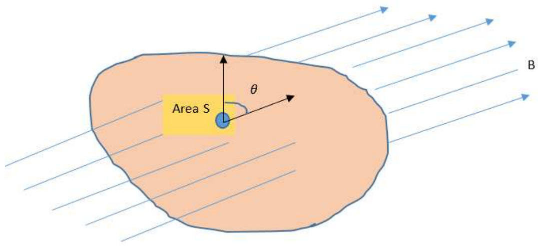
42. Gauss Law of Magnetism

Consider a small area element ΔS of a closed surface S .The magnetic flux through ΔS is defined as $\Delta\varphi_B$ = B. ΔS .We divide area element into many small area elements and calculate the individual flux through each. Then net flux φ_B is

$$\sum \quad \Delta\varphi_B = \sum \quad B.\Delta S = 0$$

Compare this with Gauss's Law of electrostatics. The flux through a closed surface in that case is

$$\sum \quad B.\Delta S = q/\epsilon_o$$



Where q is electric charge enclosed by the surface. The difference between Gauss Law of magnetism and that for electrostatics is a reflection of the fact that isolated magnetic poles (monopoles) are not known to exist. There are no sources or sinks of B: the simplest magnetic element is a dipole or a current loop. All magnetic phenomena can be explained in terms of arrangement of dipoles and /or current loops. "The net magnetic flux through any closed surface is zero".

1. Statement 1 : We cannot separate the poles of a magnet by breaking it into 2 pieces Statement 2 : Magnetic monopoles do not exist
   a) Both the statements are correct
   b) Only statement 1 is correct
   c) Only statement 2 is correct
   d) Both are wrong
2. From Gauss's Law in magnetism we can conclude that
   a) $\sum \quad B.\Delta S = q/\epsilon_o$
   b) Monopoles do not exist
   c) Electric charges exist as monopoles
   d) Current loop is the source of magnetism
3. Which of the following is true?
   a) Magnetic flux through a closed surface is zero
   b) Magnetic field lines are non-intersecting
   c) Magnetic monopoles do not exist
   d) All the above
4. The surface integral of magnetic field over a closed surface is equal to
   a) Zero b) q/ϵ_o c) total electric flux d) current loop

****

SECTION C - ANSWERS

1. (a) 0.4 Nm
2. (c) is zero, otherwise there would be a field falling as $1/r^3$ at large distances outside the toroid.
3.(d). Equal
4. (b) 327^0 C

$$\frac{x_1}{x_2} = \frac{T_2}{T_1}$$

$$\frac{x_1}{0.5\, x_1} = \frac{T_2}{273 + 27}$$

$T_2 = 600\ K = 600 - 273 = 327^0$ C

5.(b) $\frac{2}{3}$ Am $^{-1}$

Here M_1 = 8 Am^{-1} and B_1 = 0.6 T

$T_1 = 4\ K$, B_2 = 0.2 T, $T_2 = 16\ K$

Then for Paramagnetic materials, Magnetisation, M = (CB/T) from Curies law

Now in first case ,

$$M_1 = \frac{CB_1}{T_1}$$

secondly

$$M_2 = \frac{CB_2}{T_2}$$

$\frac{M_1}{M_2} = \frac{B_1}{B_2}\ x\ \frac{T_2}{T_1}$

$\frac{M_1}{M_2}$ = (0.6/0.2) x (16/4)

$$\frac{8}{M_2} = 3\ x\ 4$$

Hence $M_2 = \frac{2}{3}$ Am $^{-1}$

6.$(d) Both\ WA^{-1}m^{-1}$ and NA^{-2}

7. (d) Attract N1 strongly, N2 weakly and repel N3 weakly.

1 MARK

8. When a bar magnet of magnetic moment (M = m2L) is cut into two equal pieces transverse to its length, its (i) the pole strength remains unchanged ii) the magnetic moment is halved.

9. μ is < 1 and > 0, so magnetic material is diamagnetic.

10. Iron is a ferromagnetic substance while and copper is diamagnetic, the susceptibility of iron is much larger.

11. The magnetic field lines prefer to pass through iron than air because the permeability of iron is much larger than air.

12. Two Characteristics Properties

- High magnetic susceptibility
- High permeability

2 MARKS

13. The magnetic moment of each half bar magnet is M /2 but oppositely directed, so net magnetic moment of combination $\frac{M}{2}$ - $\frac{M}{2}$ =0 (zero).

14. According to Gauss's Law in magnetism, the net magnetic flux through a closed surface is Zero. It signifies that in magnetism, isolated monopoles do not exist.

15. M = mL

New magnetic moment is M' = m (2R)

Relation between R and L

π R = L

R = L/ π

M'= m (2L/ π)

16(A). Place a compass needle at the given point .If it stays in the North-South direction, then the magnetic field is due to earth. If the needle points along any direction other than North –South, then the field is due to some current carrying conductor. If the current is switched off, the needle will orient itself along the North-South direction.

16 (B). (a) Case-1 because for lowest potential energy configuration, magnetic moment should be parallel.

(b) Case-2 because stability is inversely proportional to potential energy.

17. The susceptibility of a paramagnetic substance is inversely proportional to the absolute temperature. X $\propto$ 1/T.

$\chi = c\frac{\mu_0}{T}$

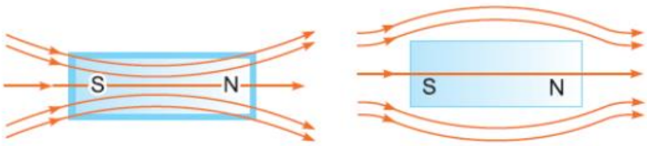
i.e. χ x T = Constant , Given T1 = 300 K

$$\frac{x_1}{x_2} = \frac{T_2}{T_1}$$

$$T_2 = \frac{x_1}{x_2} \quad T_1$$

$$= \frac{1.2 \times 10^5}{1.44 \times 10^5} \ X \, 300 = 250 \, K$$

18. This distinguishing feature is because of the difference in their relative permeability. Relative permeability is the ratio of the permeability of a specific medium to the permeability of free space. The relative permeability of diamagnetic substance is less than 1, so the magnetic lines of force do not prefer passing through the substance whereas the relative permeability of paramagnetic substance.



Paramagnetic substance (i)

Diamagnetic substance (ii)

19.The material which has relative permeabilities slightly greater one are paramagnetic whereas the material which has relative permeabilities slightly lesser than one are diamagnetic.

A ⟶ Paramagnetic

B ⟶ Diamagnetic

3 MARKS

20. (i) Direction of magnetic moment M of the current loop is perpendicular to the plane of the paper and directed inward. (ii) Torque acting on the current loop is a) maximum when M is perpendicular to B b) minimum when M is parallel to B.

21. (i) Magnetic field lines start from a North Pole and end on a South Pole. ii) Magnetic field lines form closed loops iii) Magnetic field lines do not intersect each other.

22. τ = MB sin θ

M = $\frac{\tau}{Bsin\theta}$ = $\frac{4X10^{-2}}{0.1 sinsin\, 30}$ = 0.8 JT^{-1}

23. (i) W = MB ($\cos\theta 1 - cos\theta 2$)

= 6 X 0.44 X (cos 60 – cos 90)

= 1.32J

(ii) W = 6 X 0.44 X (cos 60 – cos 180)

= 3.96 J

24. The time period of oscillation of the magnet is $T = 2\pi\sqrt{\frac{I}{MB}}$

But $I = \frac{ML^2}{12}$

When the bar magnet is cut into 2 equal pieces, perpendicular to its length, then moment of inertia of each piece of magnet about an axis perpendicular to length, passing through its centre is

$$I' = \frac{M(L/2)^2}{12X\,2} = \frac{ML^2}{12}\,\frac{1}{8}$$

Also, magnetic dipole moment M'=M/2

Its time period of oscillation is $T' = 2\pi\sqrt{\frac{I'}{M'B}} = \pi\sqrt{\frac{I}{MB}}$

25. The pole strength of a magnet depends on i) its area of cross section ii) nature of its material iii) its state of magnetisation

26. $\tau = MB \sin\theta$

Also $\tau = I\alpha$

$MB \sin\theta = I\alpha$

$MB\,\theta = I\alpha$ for small angle

$\alpha = \frac{MB\theta}{I}$

But $\alpha = \omega^2\ \theta$

So $\omega = \sqrt{\frac{MB}{I}}$

$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{MB}}$

27. Given T1 = 280 K

$H_1 = 2 \times 10^3\ Am^{-1}$

$I_1 = 4.8 \times 10^{-2}\ Am^{-1}$

Using the relation,

$x_1 = \frac{I_1}{H_1}$

$x_1 = \frac{4.8 \ x \ 10^{-2}}{2 \ x \ 103}$ = 2.4 x 10^{-5}

Now, according to the Curie's law, X ∝ 1/T.

$\frac{x_1}{x_2} = \frac{T_2}{T_1}$ T2 = 320 K

$x_2 = \frac{280}{320}$ x 2.4 x 10^{-5} = 2.1×10^{-5}

Now, the intensity of magnetisation,

$$I_2 = X_2 \, H_2$$

Since, H is independent of the temperature, i.e. $H_1 = H_2$

$I_2 =$ (2.1×10^{-5}) x (2 x 10^3) = 4.2 x $10^{-2\,2}$ Am^{-1}

5 MARKS

28. Consider a magnetic dipole (or a bar magnet) SN of length 2l having South Pole at S and North Pole at N. The strength of south and north poles are – qm and + qm respectively.

Magnetic moment of magnetic dipole m = qm 2l, its direction is from S to N.

Consider a point P on the axis of magnetic dipole at a distance r from mid-point O of dipole.

The distance of point P from N-pole, r1= (r – l).

The distance of point P from S-pole, r2= (r + l)

Let B1 and B2 be the magnetic field intensities at point P due to north and south poles respectively. The directions of magnetic field due to North Pole is away from N-pole and due to South Pole is towards the S-pole. Therefore,

$B_1 = \frac{\mu_0}{4\pi} \ \frac{q_m}{(r-l)^2}$ from N to P and

$B_2 = \frac{\mu_0}{4\pi} \ \frac{q_m}{(r+l)^2}$ from P to S

$B_1 > B_2$ and direction of resultant is from N to P and magnitude is given by

$B = B_1 - B_2 = \frac{\mu_0}{4\pi} \frac{2(qm \ 2l)r}{(r^2-l^2)2}$

But qm 2l =M magnetic dipole moment

$$B = \frac{\mu_0}{4\pi}\frac{2(M\ )r}{(r^2-l^2)2}$$

For a short dipole

$$B = \frac{\mu_0}{4\pi}\frac{2M}{r^3}$$

b. If L is the length of the wire

L =N X 2πR = N' X 2πR/2

No of turns in the new coil N' = 2N

Original magnetic moment = M = NIA = NI X πR^2

New magnetic moment M' = N' I A' = 2NI X $(\pi\ R/2)^2$

M'/M =1/2

29. If magnetic compass of dipole moment m is placed at angle θ in uniform magnetic field, and released it experiences a restoring torque

Restoring torque, τ = – magnetic force × perpendicular distance = – qmB. (2a sin θ),

τ = – mB sin θ, where qm = pole strength, m = qm.2a (magnetic moment)

Negative sign shows that restoring torque acts in the opposite direction to that of defecting torque. In equilibrium, the equation of motion

$$I\frac{d^2\theta}{dt^2} = MB\theta \text{ as } \theta \text{ is very small}$$

$$\frac{d^2\theta}{dt^2} = -\ \frac{MB}{I}\ \theta$$

$$\frac{d^2\theta}{dt^2} = -\ \omega^2\ \theta \qquad \alpha = -\omega^2\ \theta$$

It represents the simple harmonic motion with angular frequency ω

Expression for Time period:

We have, $\omega^2 = -\ \frac{MB}{I}$

But T $= \frac{2\pi}{\omega} = 2\pi\ \sqrt{I/MB}$

Assertion and Reason

30.(d) moving charge, electron, current carrying wire, naturally occurring magnet all these can create magnetic field.

31.(a) Stationary charge doesn't get affected by magnetic field, moving charge creates magnetic field.

32.(a) If $F \perp V$, at all instants then motion will be circular.

33.(b) For velocity selector the electrons can pass undeflected, when velocity and magnetic field are parallel , in that case force can be zero, both are correct but reason is not correct explanation of assertion.

34.(a) $F = q\,E + q(v \times B)$ this is called Lorentz force.

Due to electric field, acceleration $a = \frac{qE}{m}$, hence velocity will not remain constant.

35.(c) current sensitivity $I_S = \frac{\phi}{I} = \frac{NAB}{k}$

voltage sensitivity $V_S = \frac{NAB}{kR}$

$$V_s = \frac{I_S}{R}$$

36. (d) In a non-uniform magnetic field both the torque and force will act.

37. (d) Galvanometer is used to detect the direction of current in a circuit, it gives full scale deflection to the current of microampere range.

38.(c) Diamagnetic materials get repelled in magnetic fields so they exhibit magnetism but they don't have unpaired electrons so we can say that they don't have dipole moment.

39.(a) Paramagnetic materials have unpaired electrons so they exhibit magnetism.

Case Study Based Question

40. 1.(a) Diamagnetism

2.(a) Diamagnetic

3.(b) is not affected by temperature

4.(d) -1

5.(d) Copper

41. 1.(b) 60^0

2.(a) 1.5×10^{-5} Nm

3.(d) PQ_6

4.(c) $\pi : 4$

Source Based

42.1. (a) Both the statements are correct

2. (b) Monopoles do not exist

3. (d) All the above

4. (a) Zero

****